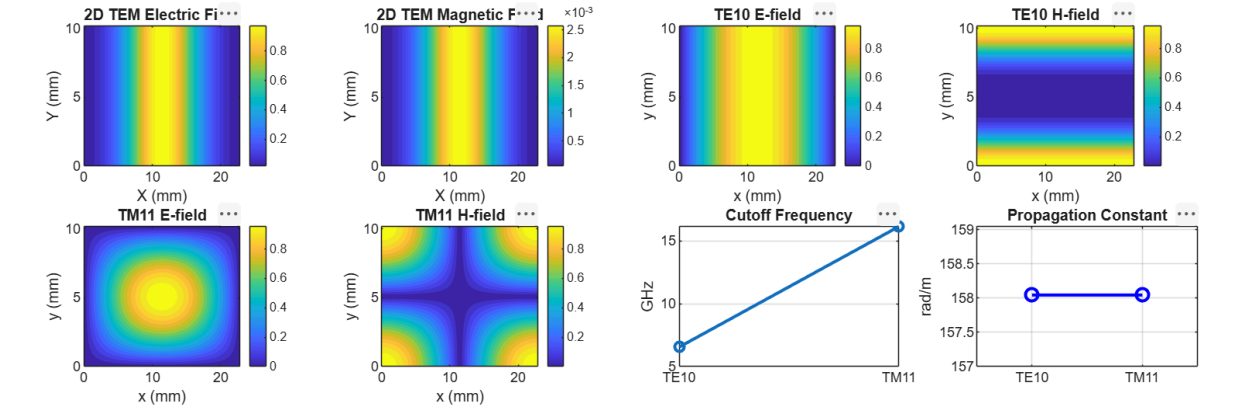
EXP 1:

Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

CODE:

```
clc;
clear;
close all;
a=22.86e-3; b=10.16e-3; c=3e8; f=10e9;
x=linspace(0,a,150); y=linspace(0,b,100); [X,Y]=meshgrid(x,y);
E=exp(-((X-a/2)/(a/4)).^2); H=E/377;
fc1=c/(2*a); fc2=c/2*sqrt((1/a)^2+(1/b)^2); k=2*pi*f/c;
beta1=sqrt(k^2-(2*pi*fc1/c)^2);
x2=linspace(0,a,100); y2=linspace(0,b,100); [X2,Y2]=meshgrid(x2,y2);
E1=sin(pi*X2/a); H1=abs(cos(pi*Y2/b)).^4;
E2=sin(pi*X2/a).*sin(pi*Y2/b); H2=cos(pi*X2/a).*cos(pi*Y2/b);
figure('Color','w');
subplot(2,4,1); contourf(X*1e3,Y*1e3,E,30,'LineColor','none'); colorbar; title('2D TEM Electric Field'); xlabel('X (mm)'); ylabel('Y (mm)');
subplot(2,4,2); contourf(X*1e3,Y*1e3,H,30,'LineColor','none'); colorbar; title('2D TEM Magnetic Field'); xlabel('X (mm)'); ylabel('Y (mm)');
subplot(2,4,3); contourf(X2*1e3,Y2*1e3,E1,20,'LineColor','none'); colorbar; title('TE10 E-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,4,4); contourf(X2*1e3,Y2*1e3,H1,20,'LineColor','none'); colorbar; title('TE10 H-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,4,5); contourf(X2*1e3,Y2*1e3,E2,20,'LineColor','none'); colorbar; title('TM11 E-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,4,6); contourf(X2*1e3,Y2*1e3,abs(H2),20,'LineColor','none'); colorbar; title('TM11 H-field'); xlabel('x (mm)'); ylabel('y (mm)');
subplot(2,4,7); plot([1 2],[fc1 fc2]/1e9,'o-','LineWidth',2); set(gca,'XTick',[1 2],'XTickLabel',{'TE10','TM11'}); ylabel('GHz'); title('Cutoff Frequency'); grid on;
subplot(2,4,8); plot([1 2],[beta1 beta1],'bo-','MarkerSize',8,'LineWidth',2); xlim([0.5 2.5]); set(gca,'XTick',[1 2],'XTickLabel',{'TE10','TM11'}); ylabel('rad/m'); title('Propagation Constant'); grid on;
fprintf('TE10: fc=%.3f GHz, beta=%.2f rad/m\n',fc1/1e9,beta1);
fprintf('TM11: fc=%.3f GHz; below cutoff at 10 GHz.\n',fc2/1e9);
fprintf('TE10 propagates; TM11 does not.\n');
```

OUTPUT:



VERIFICATION:

Exp-1

TE_{10} - $f_c = 6.562$ GHz

TM_{11} $f_c = 16.456$ GHz below cutoff freq 10 GHz

beta = 158.05 rad/m

TE_{10} Propagated

TM_{11} Below cutoff freq

Calculation

TE_{10} Mode

$$f_c = c/2a = \frac{3\times10^8}{2(22.86\times10^{-3})}$$

$$\boxed{f_c = 6.562 \text{ GHz}}$$

TM_{11} mode

$$f_c = c/2\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{m}{b}\right)^2}$$

$$= \frac{3\times10^8}{2}\sqrt{\left(\frac{1}{0.0228}\right)^2 + \left(\frac{1}{0.01016}\right)^2}$$

$$= 1.5\times10^8 \sqrt{43.74^2 + 98.425^2}$$

$$\boxed{f_c\, TM_{11} = 16.42 \text{ GHz}}$$

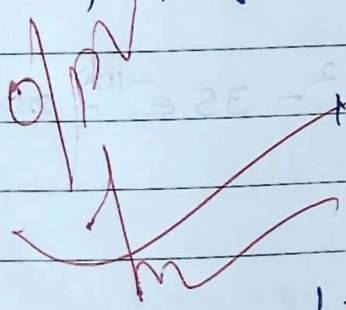
$$\beta = \frac{2\pi f}{c}\sqrt{\left(1 - \left(\frac{f_c}{f}\right)^2\right)}$$

$$= \frac{2\times3.14\times10\times10^9}{3\times10^8}\sqrt{1 - \left(\frac{6.562}{10}\right)^2}$$

$= 209.44\sqrt{0.564}$

$$\boxed{\beta = 158.087 \text{ rad/m}}$$

propagates const TM_{11} mode



$$1 - \left(\frac{f_c}{f}\right)^2 < 0$$

$$1 - \left(\frac{6.526}{10}\right)^2 < 0.$$